

EDGE AI DETAILS

1. MODEL TYPE & PURPOSE

Property	Detail
Model Type	INT8 Quantized TENet-style CNN
Task	Multi-class Audio Classification
Input	Log-Mel Spectrogram (1×98×1×104)
Output	9-class Softmax - 1×9
Data Type	INT8 (Full Integer Quantization)
Accelerator	Silicon Labs MVP v1 NPU
Purpose	On-device assistive & emergency alerts

2. DATASET SUMMARY

Property	Detail
Total Classes	9
Samples per Class	10,000
Total Samples	90,000
Class Balance	Perfectly balanced
Classes	assist, water, restroom, pain, emergency, food, noise, help, unknown
Input Shape	Log-Mel Spectrogram (1×98×1×104)
Quantization	INT8

3. MODEL ARCHITECTURE & ACCURACY

Metric	Value
Architecture	TENet-style CNN (Depthwise Separable + Residual)
Total Layers	56
Total Operations	4.9M
MAC Count	2.4M
Downsampling	98 → 49 → 25 → 13

Accuracy	~90–95%
Key Class Priority	emergency, help, pain

4. INFERENCE LATENCY

Metric	Value
Accelerator	Silicon Labs MVP v1 NPU
NPU Toolkit Version	0.3.0
Total Operations	4.9M
Accelerator Cycles	1.6M
MAC Efficiency	1.5 MACs/cycle
Unsupported Layers	0

5. MEMORY FOOTPRINT

Memory Type	Size
Flash (Model Storage)	175.2 KB
SRAM (Runtime Buffer)	42.2 KB
Temp Prepare Memory	36.5 KB
Input / Output Tensor	1×98×1×104 / 1×9 (INT8)

6. OPTIMIZATION TECHNIQUES

Techniques	Benefit
INT8 Quantization	Reduces model size & memory; speeds up inference; maintains accuracy
Depthwise Separable Convolutions	Lowers computation while preserving feature extraction
Inverted Residual Blocks (TENet-style)	Efficient feature representation with fewer parameters
Hardware Acceleration (MVP v1 NPU)	Executes fully on NPU; zero CPU fallback; deterministic latency
Balanced Dataset	Prevents class bias; improves generalization
Unknown Class Handling	Suppresses false positives from noise and out-of-vocabulary inputs